

Tao Yun Huang

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Customer Engineer · Field Service

EQUIPMENT MAINTENANCE · FAULT DIAGNOSIS · ON-SITE CUSTOMER SUPPORT

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PROFILE

Hardware engineer trained in electrical engineering, with two and a half years across two production studios delivering systems into the field — deadline-fixed public deployments for government agencies, a national museum, and major retail, where the equipment had to run on opening day with no second attempt. Owned installed systems end to end: mechanical and electrical installation, commissioning, scheduled preventive maintenance, on-site fault diagnosis and downtime recovery, remote monitoring, and direct accountability to the customer through handover and the warranty period. Strongest at the physical-signal layer — isolating noise, grounding, EMI, and bus or communication faults with instruments rather than by swapping parts. Available for shift work including nights, on-call rotation, cleanroom PPE, and travel and relocation within Taiwan.

FIELD READINESS

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|--|--|
| ✓ Shift work including night shift | ✓ 24/7 on-call rotation |
| ✓ Cleanroom work — full PPE | ✓ Travel 25% |
| ✓ Extended standing, confined-space and at-height work, lifting to 16 kg | ✓ Immediate start · relocation within Taiwan |

HARDWARE ENGINEER · IF PLUS ART 當若科技藝術 · DEC 2024 – APR 2026

- **Preventive & corrective maintenance.** Owned permanently installed electromechanical systems after handover — scheduled PM cycles, on-site fault diagnosis, downtime recovery, and remote monitoring and diagnostics over the network. Sole engineer accountable through the warranty period.
- **Fault isolation at the physical layer.** Diagnosed noise coupling, grounding problems, EMI, and bus / communication conflicts using oscilloscope, multimeter, and logic analysis — separating mechanical, electrical, and firmware root causes instead of swapping parts. Ran electromagnetic-compatibility (EMC) testing on custom hardware and closed out the interference paths it exposed.
- **Production support & quality.** Resolved technical issues arising in production, ran product verification with the manufacturing team, and took part in quality control.
- **Documentation & specifications.** Wrote technical documentation and design specifications, compiled test reports, and maintained schematics and design drawings as the controlled reference for build and service.
- **Installation & commissioning.** Installed and commissioned systems on customer sites against fixed opening dates: mechanical assembly and alignment, wiring, sensor mounting and calibration, and acceptance testing with the customer present.
- **Motion control & tuning.** Owned a 20 m single axis carrying a 500 kg moving load from drive sizing through on-site commissioning. Effective inertia changes along the travel, so closed-loop position and velocity run over an inner torque loop — one set of gains holding end to end, without overshoot, oscillation, or settling error at the end positions. Same approach applied to a 60 kg armature on an overhead-crane axis.
- **Actuation & timing.** Integrated solenoid valves and pneumatic actuators with embedded control, and synchronized mechanical motion with display and audio subsystems on a shared timing reference through continuous daily operation.
- **Customer interface.** Direct point of contact for government agencies, a national museum, and retail clients — scheduling site access, reporting equipment status, and coordinating fixes while the venue stayed open to the public.
- **Electronics build & rework.** Custom PCB schematic and layout, circuit simulation and analysis, component selection and BOM management, hand-etched boards, analog assembly, soldering and rework, wiring and power distribution, sensor-to-controller signal conditioning. Design iterations targeted lower cost and power draw and higher reliability.

Diesel-Electric Workshop

LONG-TRAVEL MOTION · ABSOLUTE POSITIONING · UNDER MAINTENANCE

National Railway Museum, Taipei · 2025

A 20 m single axis carrying six transparent screens on a 500 kg servo carriage — 600 W AZX servo through a 25:1 planetary gearhead, driven by an AZXD-SED over EtherCAT, with battery-free absolute encoder feedback and a power-off-activated brake that holds on E-stop rather than coasting. Playback is indexed to encoder position rather than to a clock, so the glass stays registered to the locomotive behind it and the error cannot accumulate. Also on site: a 60 kg locomotive armature on the overhead-crane axis. On a scheduled PM cycle since handover, with on-site fault isolation and remote monitoring.

Drive sizing · closed-loop tuning over varying inertia · commissioning · preventive maintenance & troubleshooting

AI Living Lab

SENSOR HARDWARE · CONTINUOUS DUTY

MODA · Taipei Computer Festival 2025
1,000+ users · 4 days

Custom instrumented-bicycle hardware: sensor acquisition, signal conditioning, and data normalization into a real-time processing chain. Ran continuous public duty for four days across 1,000+ users without intervention.

Sensor hardware · signal conditioning · continuous-duty reliability

Culture-Tech Exchange

NETWORKED SYSTEM · ON-SITE OPERATION
TAICCA · Pier-2, Kaohsiung · 4 days

Real-time networked system synchronizing mobile inputs to a venue display on a fixed 3-minute cycle; deployed, operated, and supported on site across the full run.

Cross-platform data sync · on-site deployment & live support

Moss Floating Layer

MULTICHANNEL AV · PERMANENT INSTALL
SCOPE TECH · Shin Kong Mitsukoshi A9 · 2025

Multichannel audio system specified, installed, and commissioned into a permanent retail environment in Taipei's busiest commercial district.

System design · installation & commissioning

TECHNICAL PROJECT MANAGER · KING ONE INTERACTIVE 王一互動科技 · JAN - DEC 2024

- **End-to-end technical delivery.** Led technical delivery of experiential technology projects across exhibitions, corporate showrooms, and trade shows — scope definition, schedule, on-site installation, and client acceptance.
- **Cross-functional coordination.** Coordinated mechanical, electrical, AV, and software scopes with internal teams and outside contractors against fixed, non-movable show-open dates.
- **Technical risk ownership.** Assessed technical risk during build and resolved integration and equipment failures on site while the show was running.
- **Delivery process.** Short iterative cycles on a Scrum-style backlog with task and issue tracking in Odoo.

EARLIER

Lecturer — Electronic Programming

PART-TIME · TEACHING
National Hsinchu Senior High School
Jul 2023 - Jun 2024

Designed and delivered an electronic-programming curriculum to a class of 13+ students — breaking technical material down for an audience with no prior background.

Technical instruction · curriculum design

Research Assistant

CONTRACT · NSTC-FUNDED PROJECT
Institute of Music, NYCU
Apr 2022 - Jan 2023

Technical support on a government-funded research project: proposal drafting and laboratory build-out, within an international collaboration with Kunsthochschule für Medien Köln (KHM). NT\$500,000–800,000 per phase.

Laboratory build-out · research support · international coordination

HANDS-ON SYSTEMS EXPERIENCE

ELECTRICAL

PCB schematic & layout · hand etching · analog circuit design · soldering & rework · wiring and harnessing · power distribution · sensor signal conditioning

MECHANICAL

Assembly and alignment · 20 m linear axis carrying a 500 kg moving load · overhead-crane axis · 3D-printed mechanism design · bench fabrication

MOTION & CONTROL

PID tuning · closed-loop position / velocity / torque · long-travel high-mass axes · stepper & servo drives over EtherCAT · absolute encoder feedback · multi-axis synchronization

PNEUMATIC & GAS	Solenoid valves · pneumatic actuators · CO ₂ / O ₂ delivery with controlled flow and valve control · leak checking by soap test and pressure decay
INSTRUMENTATION	Component-level root-cause analysis · oscilloscope · multimeter · logic analysis · EMI and noise isolation · grounding
CONTROLS & NETWORK	Embedded systems · C++ · Python · EtherCAT motion bus · UDP / OSC · network protocols · remote monitoring · Ubiquiti UniFi
NOT CLAIMED	No vacuum, plasma, or hydraulic process experience

SELECTED HARDWARE PROJECTS

Threshold · 閾 ELECTROMECHANICAL APPARATUS · 3 VERSIONS Taichung 2022 · C-LAB 2023 · NYCU 2024	Closed-loop apparatus driving a brass plate with coordinated stepper motors, solenoid valves, and pneumatic actuators. Built and rebuilt across three deployments — actuator sequencing, mechanical settling time, and the latency budget between sensing and actuation. <i>Solo — actuator control · pneumatic integration · closed-loop stability</i>
13-Voice CV Synthesizer ANALOG DESIGN · PCB FABRICATION 1 master · 12 sub-boards · 2023	Thirteen self-designed, hand-etched boards sharing a common control-voltage bus and power distribution. The engineering problem was holding voltages accurate across all boards — bus loading, crosstalk between adjacent channels, supply decoupling — diagnosed board by board on the bench. <i>Solo — circuit design · PCB layout & etching · signal-integrity debugging</i>
Circle of Confusion GAS DELIVERY · MECHANICAL DESIGN Digital Art Center, Taipei · 2024	CO ₂ / O ₂ delivery system running a controlled cyclic flow, plus a 3D-printed mechanism designed to mount on a head-worn assembly within its weight and balance limits. Delivery lines were leak-checked by soap test and pressure decay before the system ran. <i>Solo — gas delivery & valve control · leak checking · mechanical design for additive manufacturing</i>

EDUCATION

M.F.A. Multimedia & NM National Yang Ming Chiao Tung University · 國立陽明交通大學 — Institute of Multimedia NM. Research-methodology training: applying scientific method to human-machine collaboration. Thesis centered on machine-learning system design and implementation.	2020 — 2024
Electrical Engineering Feng Chia University · 逢甲大學 — Control theory, circuit design, signal processing, embedded systems	2012 — 2015

CERTIFICATIONS

UniFi Full Stack Professional (UFSP) Ubiquiti · enterprise network design, routing & switching	2025
UniFi Wireless Admin (UWA) Ubiquiti · wireless deployment & administration	2025

AVAILABILITY

Immediate start · full-time · shift and on-call rotation accepted
Open to relocation within Taiwan — Hsinchu · Taoyuan · Taichung · Tainan · Kaohsiung
Open to overseas assignment if required
Mandarin Chinese (native) · English (professional) · Japanese (basic)